Syllabus for Machine Learning

CIS 678 (Section 01, 02) — Fall 2026

This course provides a broad introduction to machine learning, focusing on how computer systems learn from data and improve their performance through experience. Topics include statistical learning, regression, classification, decision trees, ensemble methods, Bayesian learning, neural networks, genetic algorithms, explanation-based learning, reinforcement learning, and other learning frameworks. The course also introduces the applied machine learning workflow, including data preparation, model training, evaluation, hyperparameter tuning, and interpretation. Through hands-on assignments and a course project, students will implement, compare, and evaluate established machine learning methods using real-world datasets.

Contact Information:

Instructor: Dr. Yong Zhuang

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Office: MAK D-2-234

Office Hours: **Monday** 1:00 pm - 3:00 pm, MAK D-2-234
Or by appointment. Send me an email to schedule a time to meet (over Zoom or in-person).

Course Page: Course Website: https://gvsu-cis678.github.io & Blackboard

Zoom: Meeting ID: 396 668 6420, Password: 587684

Section 01:
Class time: Monday 6:00 pm - 8:50 pm (in-person/synchronous) + asynchronous
Room: DeVos Cntr Interprofess Health, Room 507 / Zoom
Midterm: (Monday) October 19, 6:00 pm – 7:50 pm
Final exam: (Monday) December 14, 6:00 pm – 7:50 pm

Section 02:
Class time: Asynchronous
Midterm: (Monday) October 19, 6:00 pm – 7:50 pm
Final exam: (Monday) December 14, 6:00 pm – 7:50 pm

Course Objectives:

After completing this course, students should be able to:

- Explain fundamental machine learning concepts and algorithms and describe their use in prediction and data-driven knowledge discovery.
- Implement and modify established machine learning algorithms and design extensions or new learning approaches.
- Compare machine learning algorithms, explain their assumptions, strengths, and limitations, and relate selected approaches to human learning processes.
- Develop end-to-end machine learning solutions involving data preparation, feature transformation, model selection, hyperparameter tuning, evaluation, and error analysis.

- Apply machine learning methods to real-world datasets and communicate experimental results through appropriate quantitative and visual analysis.
- Explain current developments and research directions in machine learning.

Prerequisites:

CIS 500 or CIS 661 or equivalent.
Recommended background: Basic programming experience and introductory knowledge of probability, statistics, and linear algebra.

Course Materials:

There are no required textbooks for this course. If you would like additional references, some good options include:

- Aurélien Géron, Hands-On Machine Learning with Scikit-Learn and PyTorch, O'Reilly Media, 2025.
- Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor, An Introduction to Statistical Learning with Applications in Python, 2023.
- Slides, recommended readings, online references, and other resources will be made available on Course website & Blackboard.

Course Delivery – Multiple Delivery:

This course is offered using a **Multiple Delivery** format. Students may choose to participate in class sessions:

- **Section 01:** Students may attend Monday class sessions in person, join synchronously through Zoom, or participate asynchronously by viewing the recordings and completing the assigned activities.
- **Section 02:** Students participate online asynchronously by viewing recorded class sessions and completing the assigned online activities.

The specific combination of delivery methods may vary by week. Regardless of how you attend, you are responsible for the same learning outcomes, assignments, and deadlines.

Technology Requirements: Students attending remotely must have reliable internet access and a device with a webcam, microphone, and speakers. All students should log in to Zoom using their official GVSU account when joining live sessions.

Blackboard and Course Communication: All announcements, assignments, submissions, grades, and recordings will be managed through Blackboard. Course materials and additional resources will be provided at Course Website. Students are expected to check Blackboard and Course Website regularly.

Attendance and Participation: Attendance and participation requirements apply to both in-person and online students. Participation may include in-class activities, Zoom engagement, or completion of assigned online activities, depending on the week.

Recording Notice: Live class sessions may be recorded for instructional purposes and made available to enrolled students. If you have concerns related to accessibility or privacy, please contact the instructor early in the semester.

Technical Support: For technical issues related to Zoom, Blackboard, or login problems, contact GVSU IT Services at it@gvsu.edu or (616) 331–2101.

Expectations: I expect the following from you to support your learning:

a) check Blackboard and Course Website on a regular basis for announcements, due dates, and submissions

b) follow posted weekly milestones and keep your project repository organized and readable

c) adhere to the CIS & GVSU policy of Academic Honesty

Blackboard: Announcements, submissions, grades, and key links will be posted to Blackboard (https://lms.gvsu.edu/). It is your responsibility to stay informed.

Note: I use Blackboard announcements to communicate time-sensitive information (for example, changes to deadlines). Ensure that your Blackboard notification settings are set up so you receive updates as they are posted.

Grading Proportions:

Course Component	Overall Weight
Quizzes and Homework assignments	30%
Project	30%
Midterm Exam	20%
Final Exam	20%
Total	**100%**

- **Homework assignments** Homework is assigned to reinforce course material, consisting of programming tasks and mathematical or written questions. Unless specified, homework should be completed individually. Sharing code or responses is not permitted. However, discussing high-level concepts and asking questions is encouraged.
- **Course project:** Students will work in groups to develop an end-to-end machine learning project using a real-world dataset. Each group must define a learning problem, prepare the data, compare at least two machine learning approaches, use an appropriate validation strategy, analyze the results, and discuss model limitations. Required deliverables include source code, a written report, and a final presentation. Individual contributions must be documented.

Grade A	Grade B	Grade C	Grades D & F
$A \geq 93\%$	$B+ \geq 87\%$	$C+ \geq 77\%$	$D+ \geq 67\%$
$A- \geq 90\%$	$B \geq 83\%$	$C \geq 73\%$	$D \geq 60\%$
	$B- \geq 80\%$	$C- \geq 70\%$	$F < 60\%$

Course Policies:

Assignments and Due Dates:

- **Due dates:** All assignments will be due at 11:59 pm Michigan time on the due date.
- **Late policy:** Assignments submitted late will incur a 10% penalty per day, capped at five days (50%). After this period, the assignment will not be accepted.
- **Accommodations:** Inform your instructor of any required accommodations to ensure successful learning. Any impediments should be discussed with the instructor.
- The instructor reserves the right to modify course policies, calendars, and due dates.
- This course adheres to the GVSU policies available at www.gvsu.edu/coursepolicies/.

Academic Honesty:

All students are expected to adhere to the academic honesty standards of Grand Valley State University. In addition, students in this course are expected to adhere to the academic honesty guidelines as set forth by the School of Computing. Details can be found at https://www.gvsu.edu/computing/academic-honesty-30.htm

I believe that you can learn a lot from your peers, both in the class and in the broader community. Therefore, I encourage collaboration with both. However, do not mistake this as a license to cheat. Learning from and with your peers is encouraged, but passing their work off as your own is prohibited. With respect to all individual assignments in this course:

- Document all collaborations.
- No electronic code transfers between students.
- Code you find online must be cited, with an active link to that code. That code should not solve the entirety of an assigned problem/project (i.e., don't have someone else do your project for you).
- You are encouraged to engage in conversations in online forums, but do not post solutions or solicit others to complete your work for you.
- You are encouraged to talk about problems with each other in non-technical terms (i.e., not code)
- Generative AI Tools: Unless an assignment states otherwise, generative AI tools may be used for explanation, brainstorming, and debugging. Any substantive use must be disclosed and cited. Students may not submit AI-generated code, analysis, or writing that they do not understand and cannot explain. Generative AI tools may not be used during quizzes or exams unless explicitly permitted. Students remain responsible for the correctness, originality, security, and academic integrity of all submitted work.
- Ultimately, you are responsible for all aspects of your submissions. You should be able to explain and defend your submission if the work is entirely your own.

Academic Resources: GVSU also provides opportunities for students to improve their academic skills through resources, such as:

- **The writing center:** The Fred Meijer Center for Writing, with locations at the Allendale and Pew/Downtown Grand Rapids campuses, is available to assist you with writing for any of your classes. For more information about these services and locations, please visit their website: http://www.gvsu.edu/wc/
- **Speech lab:** The Grand Valley Speech Lab is a peer-to-peer communication center that helps students with all elements of oral presentations. For more information about this

service, please visit their website: https://www.gvsu.edu/speechlab/

- **Research consultants:** The Center for Scholarly and Creative Excellence (CSCE) promotes a culture of active, engaged, ethical scholarship. It supports innovative faculty and student research and collaborative partnerships in the broader community. For more information, please visit their website: https://www.gvsu.edu/csce/
- **Library:** GVSU's library offers a vast collection of online resources available for students. Visit their website for more details: https://www.gvsu.edu/library/
- **Disability support resources:** If any student in this class has special needs because of a disability, please contact Disability Support Resources at http://www.gvsu.edu/dsr/ (DSR) at 616-331-2490.

Religious Observance: The university recognizes and respects religious traditions. If you require special accommodations for religious observances, inform the instructor in advance.

Emergency Procedures:

- **In Case of Emergency Fire**: Immediately proceed to the nearest exit during a fire alarm. Do not use elevators. More information is available on the University's Emergency website located at http://www.gvsu.edu/emergency.

Tentative Course Content (subject to change throughout the semester):

- **September 6 - 7, Labor Day Recess:** No classes!
- **October 26-27, Fall Break:** No classes!
- **November 25-29, Thanksgiving Recess:** No classes!

Week	Topics Covered
1	Machine learning foundations, learning paradigms, and software tools
2	Labor Day Recess (No Class)
3	Data preparation, model evaluation, and applied machine learning workflow
4	Statistical learning and regression methods
5	Classification methods
6	Decision trees and ensemble learning
7	Similarity-based and kernel learning methods
8	Midterm exam
9	Fall Break (No Class)
10	Bayesian learning methods
11	Unsupervised learning and dimensionality reduction
12	Neural networks and learning frameworks
13	Genetic algorithms and explanation-based learning
14	Reinforcement learning methods
15	Current machine learning research, Project presentations
16	Final exam